

Deep Learning and Applications: HW-2

Due date: 2025/11/25 23:59

Instructions:

1. The use of **high-level APIs** (such as Keras, Slim, or TFLearn) is **prohibited**. You are required to manually implement the forward computation. PyTorch or TensorFlow is recommended.
2. Submit your work as a single compressed file named: `HW2_StudentID.zip`. The submitted zip file must include the following:
 - **One Jupyter notebook** (.ipynb) containing your complete code and experimental results. The notebook should execute successfully from top to bottom without manual intervention.
 - **One report** (.pdf) explaining your implementation, experimental setup, results, and analysis in a clear and concise manner.

Convolutional Neural Network for Image Recognition

In this assignment, you will construct several CNN-based classifiers using a **3-category subset of the Tiny-ImageNet dataset**:

- **Labrador Retriever** (n02099712) — 400 images
- **African Elephant** (n02504458) — 400 images
- **Goldfish** (n01443537) — 200 images

Dataset

Tiny-ImageNet consists of 200 object categories, with each image resized to 64×64 .

Dataset link: <http://cs231n.stanford.edu/tiny-imagenet-200.zip>.

Your notebook must include the complete code for downloading, extracting, and preparing the dataset.

Evaluation For every trained model (Exp. 1–4), include:

- Learning curves (training/validation loss and accuracy).
- Final classification accuracy on the training and validation sets.
- Confusion matrix and at least three correct and three incorrect predictions for each class.

Task1. Design a CNN from Scratch

1. Construct your own Convolutional Neural Network (CNN) and analyze the effects of different settings such as stride size and filter/kernel size.
2. Report evaluation items.

Task2. Image preprocessing to increase Goldfish samples

Retrain the same CNN architecture from Task1 (including the 2-node latent layer).

1. Apply image preprocessing/augmentation to increase the Goldfish class from 200 to 400 images.
2. Include evaluation items.
4. Visualization of the **2-D latent features** at multiple epochs.

Task3. Zero Initialization

Repeat Task1 but initialize **all weights to zero**.

1. Analyze and compare the results with the randomly initialized CNN from Task2.
2. Include evaluation items.

Task4: ResNet-18 with Pretrained Weights

Use the enhanced dataset from Task2 and train a **ResNet-18** model with ImageNet pretrained weights.

1. Include evaluation items.
2. Provide a short discussion on the effect of pretrained features.



Figure 1: Sample images (Labrador, Elephant, Goldfish).

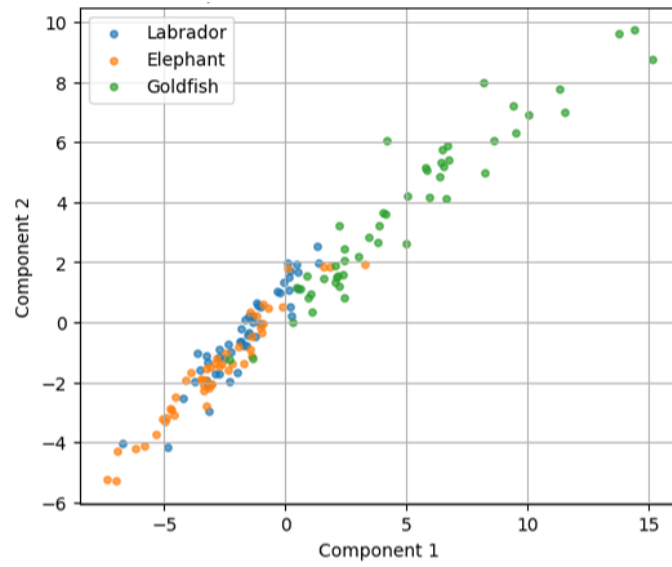


Figure 2: 2-D latent feature visualization.

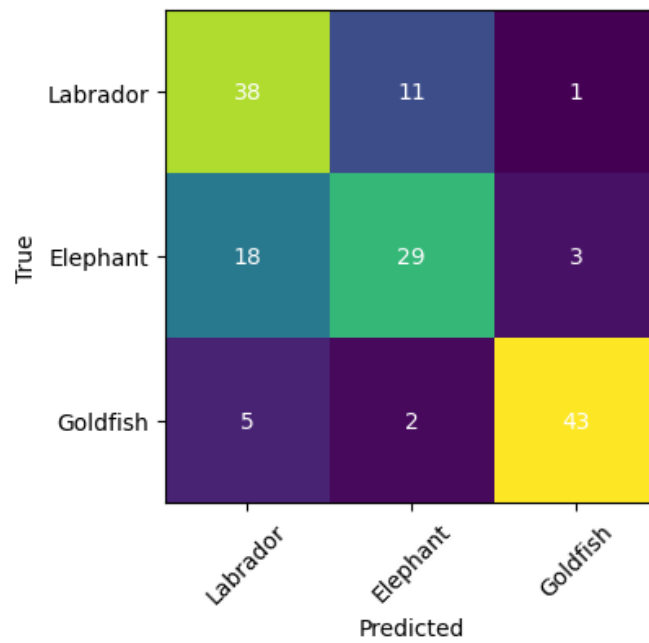


Figure 3: Confusion matrix.